MICHIGAN MEDICINE
UNIVERSITY OF MICHIGAN

Guidelines for Treatment of Prosthetic Valve Infective Endocarditis in Adults

Infectious Diseases consultation is recommended

Transition to ORAL antibiotics in endocarditis could be considered for adult patients meeting the following criteria AND requires approval from the multidisciplinary endocarditis treatment team:

Clinical status
- Clinically stable – afebrile, improving WBC
- Blood culture clearance for at least 48 – 72 hrs
- No further surgical intervention planned this admission
- Functional GI tract, able to absorb oral antibiotics
- BMI < 40

Type of endocarditis
- Blood culture with *Staphylococcus aureus, Streptococcus spp, Enterococcus faecalis,* or Coagulase-negative staphylococci, or HACEK organisms
- Organism is susceptible to an oral antibiotic listed in the treatment recommendations below, though this does not preclude use of other regimens in individual cases after discussion with ID and ID pharmacy

Logistic considerations
- **If Linezolid is used, has an appropriate linezolid trough drawn (refer to Linezolid TDM guideline)?**
- **Enrollment into Complex Outpatient Antimicrobial Therapy (COpAT) program for monitoring**
- No psychological or social issues that would prevent adherence to oral antibiotic therapy
- Plans/able follow up with ID at Michigan Medicine
- Has at least one working phone # and ideally also a secondary phone #/emergency contact, confirmed with patient
- Have a local laboratory where patient can get weekly monitoring labs
- Insurance approval for antibiotics
- If possible, deliver meds to bedside prior to patient discharge

Exclusion criteria:
- Patients with surgical indications (e.g., paravalvular abscess, metastatic sites of infections) who are not felt to be good surgical candidates

Empiric Therapy[§]	Pathogens		Subsequent Therapy (Renal Dose Adjustments May Be Necessary)	Duration of Therapy	Comments
Early (≤1 year since valve replacement): **Vancomycin** IV[4] + **Cefepime** 2 g IV q8h[1] Late (> 1 year since valve replacement): **Vancomycin** IV[4] + **Ceftriaxone** 2 g IV q24h	**Viridans group streptococci OR *Strep. gallolyticus (bovis)***	Penicillin MIC ≤ 0.12 mg/L	Preferred: **Ceftriaxone** 2 g IV q24h **OR** **Penicillin G** 4 million units IV q4h[1,2] Alternative for Severe PCN Allergy: **Vancomycin** IV[4] Preferred oral switch regimen: **Linezolid** 600 mg PO q12h Alternative oral regimen: **Amoxicillin** PO (refer to renal dosing guidelines)	6 weeks	• Consider dalbavancin in patients unable to receive recommended oral or IV antibiotic therapy. • Daptomycin should not be prescribed for Viridans group Strep. • Penicillin has a large sodium load and should be used in caution in patients with fluid restriction (dialysis, CHF, etc.). • Linezolid serum monitoring is recommended for patients on >2 weeks of therapy Amoxicillin oral regimen is a reasonable alternative, especially in patients who have baseline renal impairment; discuss dosing with ID pharmacy • Vancomycin may the most pragmatic option in dialysis patients
		Penicillin MIC > 0.12 mg/L	Preferred: **Ceftriaxone** 2 g IV q24h Alternative for Severe PCN Allergy: **Vancomycin** IV[4] Preferred oral switch regimen: **Linezolid** 600 mg PO q12h	6 weeks	
	Enterococci		See Native Valve Guideline for antimicrobial selection	≥ 6 weeks	• For Enterococci strains resistant to vancomycin, aminoglycosides, and penicillin, > 6 weeks of therapy is recommended (for all other Enterococci strains 6 weeks of therapy should be used). • Daptomycin is preferred over vancomycin for definitive outpatient therapy in most patients, except in dialysis patients • In patient with Enterococcal PV endocarditis and retained prosthetic material (e.g. graft, prosthetic valve), oral suppression following treatment should be administered when feasible

Empiric Therapy§	Pathogens	Subsequent Therapy (Renal Dose Adjustments May Be Necessary)	Duration of Therapy	Comments
	Staphylococci (MSSA or MSSE)	Preferred: **Cefazolin** 2 g IV q8h[1] OR **Oxacillin** 2 g IV q4h[2] Cefazolin allergy: **Daptomycin** 10 mg/kg IV q24h[1] Preferred oral switch regimen: **Linezolid** 600 mg PO q12h	≥ 6 weeks	• Follow baseline and weekly CK with daptomycin therapy • Cefazolin and daptomycin should be used with caution in MSSA-associated CNS infection. Oxacillin is the preferred option for patients with CNS infection. • Consider dalbavancin in patients unable to receive recommended oral or IV antibiotic therapy • Linezolid serum monitoring is recommended for patients on >2 weeks of • In patient with *S. aureus* PV endocarditis and retained prosthetic material (e.g. graft, prosthetic valve), oral suppression following treatment should be administered when feasible
	Staphylococci (MRSA or MRSE)	Preferred: **Daptomycin** 10 mg/kg IV q24h[1] OR **Vancomycin** IV[4] Preferred oral switch regimen: **Linezolid** 600 mg PO q12h	≥ 6 weeks	• Follow baseline and weekly CK with daptomycin therapy • Consider dalbavancin in patients unable to receive recommended oral or IV antibiotic therapy • If oral therapy is not chosen, Daptomycin is preferred over vancomycin for definitive outpatient therapy. However, Vancomycin may be more pragmatic in dialysis patients • Linezolid serum monitoring is recommended for patients on >2 weeks of therapy • In patient with *S. aureus* PV endocarditis and retained prosthetic material (e.g. graft, prosthetic valve), oral suppression following treatment should be administered when feasible
	HACEK Group	See Native Valve Guideline for antimicrobial selection	6 weeks	
	Candida spp.	See Native Valve Guideline for antimicrobial selection	> 6 weeks	
	Culture negative, early (≤ 1 year since valve replacement, pending definitive diagnosis)	**Daptomycin or Vancomycin** IV[4] + **Cefepime** 2 g IV q8h[1]	6 weeks	• Receipt of antibiotics prior to obtaining cultures is the most common cause of culture-negative IE. There are many infectious and non-infectious causes. An evaluation of epidemiological factors, history of prior cardiovascular infections, exposure to antimicrobials, clinical course, severity, and extracardiac sites of infection should be performed to help guide diagnosis and treatment.
	Culture negative, late (> 1 year since valve replacement, pending definitive diagnosis)	**Daptomycin or Vancomycin** IV[4] + **Ceftriaxone** 2 g IV q24h	6 weeks	

§ Prior to confirmation of pathogen

1. Refer to Antimicrobial Dosing Recommendations for dose adjustments in renal dysfunction
2. If candidate for outpatient therapy, may consider administration via continuous infusion (same daily dose)
3. Please refer to the Vancomycin Nomogram for guidance on vancomycin dosing and monitoring.
4. The risk of gentamicin and rifampin toxicity outweighs potential benefits for routine use in endocarditis patients based on current primary literature. The 2015 IDSA endocarditis guidelines recommend rifampin and/or gentamicin for some patients with endocarditis, which pre-date literature evaluating the efficacy and safety of these therapies

Last revised: 7/2025	**Antimicrobial Stewardship Subcommittee:** 7/2025	**P&T:** 12/2025

References

	Citation
1	Iversen K, Ihlemann N, Gill SU, et al. Partial Oral versus Intravenous Antibiotic Treatment of Endocarditis. *N Engl J Med*. 2019;380(5):415-424.
2	Freling S, Wald-Dickler N, Banerjee J, et al. Real-World Application of Oral Therapy for Infective Endocarditis: A Multicenter, Retrospective, Cohort Study. *Clin Infect Dis*. 2023;77(5):672-679.

Revision History

Date	Revision
03/2021	Updated vancomycin dosing & hyperlinks
01/2024	Updated oral step-down options
11/2024	Revisions throughout
07/2025	Updated and revised oral switch options

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time-to-time change. The authors of these guidelines have made all attempts to ensure accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source